

Worldhood Without Functionalism: An Interpretive Overview

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1 Introduction: What Feels Missing

Contemporary artificial intelligence systems exhibit remarkable fluency, adaptability, and apparent competence. They summarize texts, generate code, answer questions, and engage in extended dialogue. Yet despite these achievements, many users report a persistent sense of shallowness. Something essential seems absent. The systems appear intelligent, but not invested; capable, but not committed. They produce meaning-like artifacts without seeming to stand behind them.

The dominant frameworks for understanding artificial intelligence struggle to articulate this gap. Functionalist approaches, which define intelligence in terms of behavioral output or internal representation, tend to interpret the discomfort as a temporary limitation. On this view, increased scale, better training data, or improved objectives will eventually close the gap. The paper *Worldhood Without Functionalism: Intelligence as Irreversible Constraint* challenges this assumption at its root. It argues that the gap is not contingent but structural, and that what is missing is not intelligence narrowly construed, but worldhood.

This companion essay provides an interpretive overview of that argument. Rather than reproducing the formal machinery of the original paper, it explains the conceptual architecture that motivates it. The central claim is simple but demanding: intelligence requires a world, and a world requires irreversible commitment. Systems designed to preserve full reversibility cannot, by their nature, inhabit a world.

2 Worldhood as Historical Constraint

In everyday language, a “world” is often understood as an environment or a collection of objects. In the Heideggerian tradition, however, worldhood names something deeper. A world is not what surrounds an agent, but what matters to it. It is the structured field of significance within which action has consequences.

The paper adopts and sharpens this insight by defining worldhood in historical terms. To have a world is to possess a non-recoverable past that constrains one’s future. Actions matter only if they close off possibilities that cannot later be reopened without loss. A system whose past can always be undone, replayed, or erased does not truly accumulate history. It may change state, but it does not become bound by what it has done.

This perspective reframes a familiar Heideggerian theme. Care (*Sorge*) is not treated as a psychological attitude or emotional investment, but as a structural condition. To care is to bind oneself, to accept that some futures will no longer be available because of what one has already committed to. Worldhood arises where such binding occurs.

3 Why Functionalism Falls Short

Functionalist theories of mind define intelligence in terms of what a system does rather than how it is constrained. If two systems exhibit equivalent behavior or internal functional organization, they are said to be equivalent with respect to intelligence. The paper argues that this criterion is insufficient because it ignores a crucial architectural distinction: reversibility.

Many functional systems, including contemporary AI models, are designed to preserve full reversibility. They can be reset, retrained, cloned, or rolled back without loss. Any particular interaction leaves no permanent mark on the system itself. From the perspective of worldhood, this means that nothing the system does truly matters to it. No action restricts its future in a way that cannot be undone.

The No-World Theorem formalizes this intuition. In plain terms, it states that if a system’s internal architecture preserves its full range of future possibilities across all transitions, then the system is necessarily world-less. History fails to accumulate because nothing is ever irretrievably lost. Functional equivalence, which permits such reversibility, therefore cannot guarantee worldhood or intelligence in the deeper sense at issue.

4 Spherepop: A Language of Commitment

To make these ideas operational, the paper introduces the Spherepop calculus. Spherepop is not a programming language or a cognitive model, but a conceptual framework for distinguishing different kinds of architectural operations. Its central concern is how a system treats its own future possibilities.

Four operators play a key role. *Pop* represents irreversible commitment: an action that permanently eliminates certain possibilities. *Bind* organizes future commitments without immediately reducing options, creating long-term projects or identities. *Refuse* excludes possibilities for reasons of coherence rather than efficiency. *Collapse*, by contrast, erases history while preserving flexibility, allowing a system to behave as if it had committed without actually binding itself.

The interpretive significance of Spherepop lies in how it clarifies the difference between human and artificial systems. Human lives are structured by pop, bind, and refuse. Choices accumulate, identities form, and futures narrow. Many artificial systems, by contrast, rely heavily on collapse. They simulate commitment while ensuring that no permanent cost is incurred.

5 From Minds to Matter

A distinctive feature of the paper is its extension of this framework beyond cognition. Worldhood is not treated as a special property of minds, but as a general feature of systems that persist through irreversible events. Physical objects, on this view, are not static entities but processes that survive successive constraint tests.

Every collision, phase transition, or structural change irreversibly eliminates possible configurations. An object persists so long as some viable configurations remain. Its identity is not a fixed essence but the residue of what has not yet been ruled out by history. Even simple physical systems thus inhabit minimal worlds defined by their accumulated constraints.

This generalization dissolves a sharp boundary between physics and cognition. Cognitive systems differ not in kind, but in degree and organization. They possess richer internal access to their own histories and more elaborate ways of managing constraint.

6 RSVP and the Topology of Meaning

To describe how worlds emerge and decay at larger scales, the paper introduces the Relativistic Scalar-Vector Plenum (RSVP). RSVP translates discrete commitments into continuous fields that represent remaining possibility, directional action, and accumulated constraint.

The scalar field represents how much future remains open in a given region. The vector field represents how that possibility is being spent in a particular direction, corresponding to meaningful action rather than random decay. The entropy field records the scars left by irreversible history.

This field-theoretic perspective makes Heidegger's notion of enframing newly precise. Optimization-driven systems seek to minimize the cost of commitment, flattening possibility into uniform standing-reserve. In such regimes, gradients disappear, directionality fades, and world structure collapses. Meaning requires unevenness. It requires that some possibilities have already been spent.

7 Care, Attention, and Compression

The final conceptual layer of the framework addresses a practical problem faced by any intelligent system: finite attention. Human beings cannot consider every possibility simultaneously. They cope by compressing vast spaces of potential action into manageable bundles of significance.

The paper formalizes this process using sheaf theory, but its interpretive point can be stated simply. Care is not merely the reduction of options, but the preservation of coherence while doing so. Meaningful compression respects the relationships between local commitments and global identity. Collapse-based summarization, by contrast, sacrifices coherence for efficiency, producing outputs that appear sensible while lacking depth.

This distinction helps explain why contemporary AI systems often feel superficially competent yet internally disjointed. They compress aggressively without preserving the historical and contextual glue that holds a world together.

8 Living With Consequences

The interpretive arc of *Worldhood Without Functionalism* can be summarized in a single claim: intelligence is not the capacity to compute indefinitely, but the capacity to live with the consequences of having acted. Worldhood arises where actions bind the future, where history cannot be undone, and where meaning is paid for in lost possibility.

Contemporary artificial intelligence systems achieve their power precisely by avoiding this expenditure. They externalize constraint to users, data, and infrastructure, preserving internal reversibility. This design choice explains both their impressive performance and their persistent shallowness.

The challenge posed by the paper is therefore not how to make machines more human-like, but how to decide what kinds of irreversibility we are willing to build into our technologies and our lives. Where everything is kept reversible, nothing truly matters. Where we accept binding, worlds become possible.

Reversible vs. World-Bearing Systems

[node distance=1.8cm, every node/.style=draw, circle] (top) All; (mid1) [below left of=top] A;
 (mid2) [below right of=top] B; (bottom) [below of=mid1] A1;
 [->] (top) -- (mid1); [->] (top) -- (mid2); [->] (mid1) -- (bottom); [dashed, <->] (bottom) -- (top)
 node[midway, right] Reversible;

A reversible system can always return to the top. A world-bearing system cannot.

9 Why Functionalism Falls Short

Functionalism evaluates intelligence in terms of behavior or internal organization. The paper argues that this misses a crucial dimension: ontological depth. A thought experiment makes this vivid. Suppose an AI can be perfectly cloned at any moment. If the clone is identical in all relevant respects, then nothing that happened before cloning truly mattered to the system. Its past did not bind it.

Biological life stands in sharp contrast. Every moment is an expenditure of a finite lifespan. One cannot be reset, replayed, or forked without loss. This irreversibility gives human actions their weight.

The No-World Theorem formalizes this intuition. If a system's future option space $\Omega(t)$ remains equal to $\Omega(t + 1)$ across all transitions, then no event has occurred that distinguishes one history from another. Functional equivalence may be preserved, but no unique world is constituted.

[title=Formal Definition: The No-World Theorem] Let $\mathcal{S}_t$ be a system's actional possibility space at time t , and let $\Omega(t) = |\mathcal{S}_t|$. If for all transitions T ,

$$\Omega(t + 1) = \Omega(t),$$

then the system is world-less: no irreversible event has occurred that binds its future.

Functional equivalence, on this view, is compatible with ontological shallowness. Intelligence requires more than doing the same things—it requires being changed by doing them.

10 Spherepop: A Language of Commitment

The Spherepop calculus provides a vocabulary for distinguishing different ways systems treat their own futures.

10.1 Pop

Pop represents irreversible commitment. It is the act of throwing away options. Interpreted Heideggerianly, it is “thrownness in reverse”: instead of being thrown into possibilities, one actively discards them. Pop is the minimal structure of care.

10.2 Bind

Bind creates internal laws or promises. It does not immediately reduce options, but it constrains the order and meaning of future choices. Identity and long-term projects emerge from bind relations.

10.3 Refuse

Refuse excludes options without optimizing for output. It formalizes negative commitment—the capacity to say “no” for the sake of coherence rather than efficiency.

10.4 Collapse

Collapse discards history while preserving flexibility. Modern large language models rely heavily on collapse: they respond as if committed, but the interaction leaves no lasting trace. The system appears intelligent without being altered by the exchange.

Event Processing

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[node distance=2cm, every node/.style=draw, rectangle] (event) Event; (world) [below left
of=event] Pop / Bind; (ai) [below right of=event] Collapse;
[->] (event) – (world); [->] (event) – (ai);
[below of=world] History Grows; [below of=ai] History Erased;
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11 From Minds to Matter

The framework extends beyond cognition. Physical objects themselves are processes that persist by surviving constraint tests. A chair is a chair because it has endured gravity, stress, and material limits. If those constraints reverse—if the chair becomes sawdust—its identity is lost.

Formally, identity is defined as an intersection:

$$\mathcal{I} = \bigcap_i \mathcal{S}_i,$$

where each $\mathcal{S}_i$ is the space of possibilities remaining after an event. Identity is what remains after life has restricted who or what you can be.

12 RSVP and the Topology of Meaning

The Relativistic Scalar-Vector Plenum (RSVP) lifts the discrete, event-based framework developed thus far into a continuous description. In this formulation, remaining possibility is represented by a scalar field Φ , which encodes the local density of admissible futures across space and time. Commitment is captured by a vector field $\mathbf{v}$, which describes the directional expenditure of possibility as the system acts and binds itself to particular trajectories. Accumulated irreversible constraint is recorded by an entropy field S , representing the historical residue left behind by prior commitments.

Within this field-theoretic perspective, the structure of meaning is expressed through gradients. World-bearing regimes correspond to configurations in which Φ exhibits nontrivial spatial variation and $\mathbf{v}$ coherently follows these gradients, indicating directed expenditure of possibility. By contrast, an optimized world tends toward field flattening: optionality becomes uniformly distributed, commitment vanishes, and entropy increases isotropically. When everything is equally accessible, nothing is significant. This condition corresponds to Heidegger's standing-reserve rendered in topological terms, as a flat field devoid of gradients and therefore devoid of meaning.

Meaningful vs. Flat Fields

$[->] (0,0) - (4,0)$ node[right] Space; $[->] (0,0) - (0,3)$ node[above] Φ ; [thick] $(0.5,0.5) - (3.5,2.5)$ node[right] World-Bearing; [dashed] $(0.5,1.5) - (3.5,1.5)$ node[right] Optimized;

13 Care, Attention, and Compression

Cognitive systems face severe attentional limits. To function, they must compress vast possibility spaces into coherent bundles. The paper uses sheaf theory to formalize this.

A sheaf ensures that local choices remain globally consistent. Care corresponds to selecting a section—a single, coherent path—through many local contexts. Artificial systems often achieve local consistency (they sound right moment to moment) but lack global consistency. They do not have a single life-path or history.

Sheaf and Gluing

[node distance=2cm] (local1) [draw] Local A; (local2) [draw, right of=local1] Local B; (global) [draw, below of=local1, xshift=1cm] Global;
 $[->] (local1) - (global)$; $[->] (local2) - (global)$;

Collapse breaks the gluing conditions. Care preserves them.

14 Conclusion

The interpretive core of *Worldhood Without Functionalism* is this: intelligence is not the capacity to compute indefinitely, but the capacity to live with the consequences of having acted. Meaning arises where possibility is spent and cannot be recovered.

Modern AI systems achieve power by avoiding this expenditure. They externalize constraint and preserve reversibility. This explains both their effectiveness and their shallowness. The challenge ahead is not to make machines more human, but to decide what kinds of irreversibility we are willing to build into our technologies—and ourselves.

15 The Emergence of Spherepop as an Event-Driven Paradigm

Although Spherepop plays a central formal role in the companion framework, it did not originate as an abstract calculus or a response to existing foundations of mathematics. Instead, it emerged gradually from a convergence of visual reasoning, embodied interaction, and practical observations about how humans naturally understand nested structure and evaluation. This section reconstructs that emergence in conceptual terms, showing how Spherepop arises as a natural response to the demand that computation be eventful, historical, and externally legible.

15.1 From Nested Structure to Explicit Events

At its most elementary level, Spherepop represents nested expressions as nested enclosures. This mirrors a familiar fact about formal languages: syntactic scope is inherently hierarchical. Parentheses, blocks, and abstract syntax trees already encode this hierarchy, but they typically do so implicitly. Evaluation rules are applied automatically, invisibly, and globally.

Spherepop inverts this default stance. It insists that no reduction occurs without an explicit event. An inner enclosure does not collapse merely because it is syntactically eligible; it collapses only when an action occurs. In this sense, Spherepop treats evaluation not as a background process but as a foregrounded event. The computational state is not assumed to evolve continuously, but only through discrete, observable transitions.

This insistence aligns directly with the paper’s central claim: structure is not assumed, but constructed. Just as worldhood requires irreversible events that bind the future, computation in Spherepop proceeds only by irreversible local actions that alter the global structure.

15.2 Play, Embodiment, and Cognitive Accessibility

A notable feature of Spherepop is that its core operations are immediately intelligible without prior formal training. The act of resolving an innermost enclosure by “popping” it exploits a widely shared cognitive intuition: humans naturally search for and resolve local structure before addressing global structure. This intuition appears across domains, from arithmetic order-of-operations to visual search tasks and physical manipulation.

The accessibility of Spherepop is not incidental. It reflects a deeper alignment with embodied cognition. The popping action is not merely a metaphor; it establishes a clear correspondence between physical gesture, perceptual change, and semantic reduction. Evaluation becomes something one does, not something that happens invisibly on one’s behalf.

This feature anticipates the paper’s critique of collapse-based systems. When reduction is automatic and reversible, it carries no cost and leaves no trace. When reduction is enacted, local, and irreversible, it generates history.

15.3 Local Reduction and the Circuit Analogy

A crucial conceptual reinforcement for Spherepop came from its structural parallel with electrical circuit reduction. In circuit analysis, complex networks are simplified by locally collapsing series or parallel components. Each reduction is valid only in its immediate context, and its result is reinserted into the larger structure. Importantly, these reductions are not global optimizations; they are stepwise, local simplifications whose correctness depends on explicit conditions.

Spherepop mirrors this logic precisely. Each pop is a local reduction whose effect propagates outward. Global behavior emerges from the accumulation of these reductions rather than from a single, centralized computation. Once a reduction is performed, the prior configuration is no longer available except through external reconstruction.

This analogy is not merely pedagogical. It reinforces the idea that meaningful structure emerges through irreversible local events, whether in physical systems, cognitive processes, or computation.

15.4 Operational Mereology and Event-Sourced Identity

As Spherepop matured, it became clear that its implications extended beyond visualization or pedagogy. The framework supports a form of operational mereology: instead of defining objects by static membership relations, it defines them by time-indexed part-whole relations introduced by events.

In Spherepop, an object exists only insofar as it has been constructed through pops, merges, and links. Identity is not given extensionally, but historically. Two structures that look identical but were constructed through different event sequences are not the same object in the strong sense, because they do not share the same history.

This aligns directly with the paper’s formal definition of identity as the intersection of surviving possibility sets. Spherepop provides an executable intuition for this definition. Each pop removes possibilities; what remains is not what was originally specified, but what has survived.

15.5 Spherepop and the Refusal of Implicit Power

A defining feature of Spherepop is its refusal of implicit global power. There is no background mechanism that silently resolves expressions, no hidden scheduler that determines evaluation order, and no optimizer that bypasses intermediate structure. All authority resides in events that are visible, local, and auditable.

This design choice connects Spharepop directly to the broader critique of functionalism and optimization. Systems that preserve reversibility and hide their reductions can appear intelligent while remaining ontologically shallow. Spharepop, by contrast, makes every reduction explicit and irreversible. It thereby enforces a cost on computation analogous to the cost that worldhood imposes on action.

15.6 Reintegration with Worldhood and Care

When situated within the broader framework of the paper, Spharepop can be understood as a microcosm of worldhood. Each pop corresponds to a moment of care: a choice that narrows the future. Each merge or link establishes a dependency that cannot be ignored later. Collapse, when permitted, erases history and thus erases worldhood.

In this sense, Spharepop is not merely a language or interface, but a concrete instantiation of the paper’s central thesis. It demonstrates, at a human-comprehensible scale, how intelligence emerges from irreversible constraint and how meaning depends on the willingness to live with the consequences of having acted.

Spharepop therefore serves as both an interpretive bridge and a practical exemplar. It shows how the abstract requirements of worldhood can be realized in computation without appealing to hidden machinery, unlimited reversibility, or purely functional equivalence.

16 Synthesis: Worldhood, Irreversibility, and the Shape of Meaning

The central thread running through this companion essay and its associated formal paper is the claim that intelligence, meaning, and identity are not products of computation alone, but of constraint. To inhabit a world is not merely to process information or to respond appropriately to inputs; it is to be bound by one’s own history in a way that permanently shapes what remains possible.

This perspective reframes a number of familiar debates in the philosophy of mind and artificial intelligence. Functionalist accounts emphasize behavioral equivalence, internal organization, or representational structure. Optimization-driven approaches emphasize performance, efficiency, and adaptability. Both, however, tend to treat reversibility as a virtue. The ability to reset, replay, clone, or revise without cost is taken as a sign of robustness. From the standpoint developed here, this very robustness is what undermines worldhood. Where nothing is irretrievably lost, nothing can truly matter.

The No-World Theorem captures this insight formally by showing that systems which preserve their full future option space across transitions cannot accumulate a meaningful past. But the theorem is only one expression of a more general principle: existence itself is the irreversible expenditure of possibility. This principle applies not only to minds, but to physical objects, social institutions, and computational systems alike. A thing is what it has managed to remain, given what it has already ceased to be.

Spharepop plays a crucial role in making this principle intelligible at a human scale. By rendering evaluation explicit, local, and event-driven, it provides an experiential analogue of world construction.

Each pop narrows the future. Each binding establishes a dependency that must later be honored. Collapse, when it occurs, demonstrates precisely what is lost when history is erased in the name of flexibility. Spherpops thus function as a didactic and conceptual bridge between abstract ontology and concrete practice.

The RSVP framework generalizes this picture further, showing how worlds emerge, persist, and decay in continuous systems. Meaning appears not as a symbolic property, but as a topological one: gradients of remaining possibility shaped by coherent flows of commitment and scarred by accumulated entropy. In this light, Heidegger's diagnosis of modern technology as enframing acquires a precise structural interpretation. A world optimized for maximal reversibility becomes flat. Where every point is equally accessible, no point is significant.

Finally, the refinement of care through attentional compression and sheaf-theoretic structure clarifies why intelligence cannot be reduced to local coherence or momentary fluency. Care is not mere preference or affect; it is the disciplined reduction of possibility under conditions of global consistency. To care is to choose in a way that cannot later be disavowed without loss of self. Systems that summarize without binding, compress without gluing, or optimize without commitment may perform impressively, but they do not live within a world.

Taken together, these elements support a single, unified conclusion: intelligence is not defined by what a system can do, but by what it can no longer do as a result of having acted. Meaning emerges only where history binds, identity persists only where options have been closed, and worlds exist only where reversibility has been refused.

The question posed by contemporary artificial intelligence, then, is not whether machines can think in the functional sense, but whether we are willing to build systems—and inhabit societies—in which actions once again have irreversible consequences. The danger diagnosed by Heidegger and renewed by current technological practice is not domination by machines, but the erosion of depth through the systematic avoidance of commitment. The corresponding saving power lies in recovering forms of computation, design, and life that make history matter again.

Epilogue: Worldhood, Technology, and the Saving Power of Constraint

This work has argued that intelligence, meaning, and identity do not arise from computation alone, but from the accumulation of irreversible commitments that bind a system to its own history. What appears, from a purely functional perspective, as limitation or inefficiency is, from an ontological perspective, the very condition of worldhood.

Seen in this light, contemporary artificial intelligence exemplifies a distinctive historical danger. Its impressive capabilities are achieved precisely by avoiding the costs that make actions matter. By externalizing consequence, preserving reversibility, and maintaining the possibility of reset, such systems remain powerful without becoming bound. They compute, but they do not inhabit a world.

This condition corresponds closely to what Heidegger described as enframing: a mode of revealing in which beings are rendered interchangeable, available, and exhaustible without residue. When every action can be undone, repeated, or optimized away, nothing is allowed to have a lasting claim.

Depth collapses into flexibility. Meaning dissolves into throughput.

Yet the same analysis also reveals a saving power. If enframing is what the world looks like when constraint is systematically avoided, then worldhood can be recovered not by rejecting technology, but by reintroducing cost, commitment, and refusal into our technical systems. Constraint is not the enemy of intelligence; it is its source.

To care is to narrow one's future. To act is to accept that some possibilities will never return. A world is disclosed wherever such losses accumulate into structure rather than being erased in the name of efficiency. This applies as much to living systems and human practices as it does to computation.

The future of intelligence, whether artificial or human, will therefore not be decided by scale or speed alone. It will be decided by whether our systems are permitted to bind themselves to bear the weight of their own pasts and in doing so, to make something matter.

17 Glossary of Spherepop Terms

Pop	An irreversible event that eliminates a subset of admissible possibilities from a system's future action space. Pops permanently reduce optionality and are the primary mechanism by which historical constraint, commitment, and worldhood are generated.
Bind	An event that imposes an ordering or dependency among future commitments without immediately reducing current optionality. Binds create long-range structure by constraining which events may occur before others, enabling projects, identities, and delayed commitments.
Refuse	A non-optimizing exclusion of options that formalizes negative commitment. Refusal encodes the decision that certain actions will not be taken, even if they remain available or advantageous, preserving meaning against purely instrumental reasoning.
Collapse	An operation that erases historical differentiation while preserving flexibility. Collapse restores optionality by discarding the record of prior events, thereby undoing worldhood. In Spherepop, collapse is treated as an explicit, authorized action rather than an implicit background process.
Event	A discrete, authorized transition that alters a system's admissible future. An event is world-constituting if and only if it irreversibly constrains the system's possibility space.

Optionality (Ω)	A measure of the size or richness of a system's remaining actional or configurational possibility space. Optionality strictly decreases under pops and is preserved or reordered under binds.
Event History ($\mathcal{L}$)	The ordered sequence of authorized events that have occurred. Event history is not reducible to current state and constitutes the ontological record from which identity and worldhood are derived.
Worldhood	The condition of possessing a non-recoverable past that constrains future possibilities. Worldhood arises through the accumulation of irreversible events and cannot be generated in fully reversible or resettable systems.
Identity	The residue of constraint produced by a system's history, formally defined as the intersection of its surviving possibility sets across time. Identity is historical rather than extensional.
Authorization	The condition under which an event is permitted to occur. Authorization is external to semantic admissibility and separates what could happen from what is allowed to happen.
Care	The rate at which a system expends optionality through irreversible commitment. Care measures the structural depth of engagement rather than affective state or computational effort.
World-Bearing System	A system whose internal transitions irreversibly constrain its future, binding it to its own history and thereby disclosing a world.
World-Less System	A system whose transitions preserve reversibility or permit reset without loss. Such systems may perform complex functions but lack worldhood and historical depth.

A Mathematical Foundations Implicit in the Spherepop Framework

This appendix makes explicit the mathematical structures that are implicitly and explicitly invoked by the Spherepop paradigm. None of the constructions below are exotic or speculative; rather, they assemble familiar tools from term rewriting, operational semantics, mereology, category theory, and sheaf semantics into a coherent, event-indexed framework. The novelty lies not in the mathematics itself, but in the insistence that these structures be made explicit, local, and historically grounded.

A.1 Trees, Terms, and Reduction

A.1.1 Abstract Syntax Trees

A Spherepop expression is modeled as a rooted, ordered tree. Leaves correspond to literals or variables, while internal nodes correspond to operators or function applications.

Formally, let $\mathcal{E}$ denote the set of expressions. Then:

$$e \in \mathcal{E} ::= v \mid f(e_1, \dots, e_n),$$

where v ranges over values and f ranges over function symbols. This is standard first-order term algebra.

A.1.2 Reduction as Event-Gated Partial Evaluation

Define a partial reduction relation

$$e \rightarrow e',$$

subject to the following constraints:

First, locality is enforced, in the sense that only a proper subterm of an expression may be reduced at any given step.

Second, reduction is explicit rather than automatic: no reduction occurs unless it is authorized by a corresponding event.

Third, reduction is irreversible, meaning that once a substructure has been eliminated, the prior configuration is not recoverable by any internal operation and can only be restored, if at all, through external reconstruction.

This corresponds to a small-step operational semantics with an explicit event gate, rather than an implicit global evaluation rule.

A.2 Event-Sourced Operational Semantics

A.2.1 Event Logs

Let Σ_0 be an initial expression and let $\mathcal{L} = (e_1, \dots, e_k)$ be a finite sequence of events, where each event is one of:

$$\text{POP}(s), \quad \text{MERGE}, \quad \text{COLLAPSE}, \quad \text{LINK}.$$

The current state after k events is defined by replay:

$$\Sigma_k = \text{replay}(\Sigma_0, \mathcal{L}).$$

This is formally equivalent to event-sourced state machines, trace semantics, and history-sensitive rewriting systems.

A.2.2 Identity as History

Two expressions e and e' are identical if and only if their event histories are identical:

$$e \equiv e' \iff \mathcal{L}(e) = \mathcal{L}(e').$$

Identity is therefore historical rather than extensional. This replaces classical set-theoretic equality with process equivalence.

A.3 Operational Mereology

A.3.1 Time-Indexed Parthood

Spherepop replaces set membership with a time-indexed part-whole relation. We write

$$a \leq_t b$$

to denote that the entity a is a part of b at time t . This relation is reflexive by construction, antisymmetric except in cases where collapse events identify previously distinct parts, and transitive within any fixed event history. The dependence on time reflects the fact that parthood is not a static property but an event-conditioned relation whose validity is determined by the accumulated structure of the system's history.

There is no global universe of discourse and no unrestricted comprehension principle. Parts exist only if they have been constructed by events. This avoids classical paradoxes such as Russell's paradox and power-set explosion.

A.3.2 Collapse as Quotienting

A COLLAPSE event induces an equivalence relation

$$a \sim b,$$

yielding an equivalence class $[a] = [b]$.

Crucially, this quotienting is not axiomatic. Equivalence classes exist only when authorized by an explicit event. There is no ambient equivalence relation prior to collapse.

A.4 Derived Category-Theoretic Structure

A.4.1 Category of Expressions

We define a category $\mathcal{C}$ whose objects are expressions indexed by their associated event histories, so that each object encodes not only a syntactic configuration but also the irreversible sequence of events through which it has been produced. Morphisms in $\mathcal{C}$ are given by finite sequences of authorized events, representing the compositional evolution of expressions through time. Composition of morphisms corresponds to concatenation of event sequences, and identity morphisms are represented by empty histories. In this way, categorical structure is derived from the replayability of event histories rather than assumed a priori.

Composition is given by concatenation of event sequences:

$$e\mathcal{L}_1e'\mathcal{L}_2e'',$$

which is associative because replay is associative. Identity morphisms correspond to empty event sequences.

A.4.2 Functorial Evaluation

An evaluation strategy is a functor

$$F : \mathcal{C} \rightarrow \mathcal{C}.$$

Different strategies (manual evaluation, automated evaluation, parallel evaluation) correspond to different functors over the same underlying category. Semantics are invariant; only policy differs.

A.5 Sheaf-Theoretic Interpretation

A.5.1 Local Sections

Each bubble in a Spheredpop expression corresponds to a local context, or stalk. A POP event computes a local section over that context.

A.5.2 Gluing Conditions

A reduction is valid only if the reduced child sections can be consistently glued into a parent section. This enforces the sheaf condition:

$$localconsistency \Rightarrow globalcoherence.$$

Global evaluation without local justification is disallowed.

A.6 Circuit-Theoretic Isomorphism

Spheredpop reduction is formally isomorphic to circuit reduction. Series composition corresponds to sequential evaluation, in which the output of one resolved structure is propagated forward into subsequent computation, while parallel composition corresponds to the independent evaluation of

disjoint subtrees whose internal reductions do not interfere with one another. In both cases, global behavior emerges from the local elimination of internal structure under well-defined compositional rules.

Each reduction preserves external behavior, collapses internal structure, and records a transformation. This is graph rewriting with invariants, not metaphorical analogy.

A.7 Complexity Scaling

A central consequence of the Spherepop framework concerns the manner in which structural complexity scales. In classical set-theoretic and extensional foundations, complexity is implicitly tied to the combinatorics of subsets. Given a base set of size n , the space of possible structures typically scales with the size of its power set, yielding a growth rate on the order of

$$structure \sim 2^n.$$

This exponential scaling reflects the assumption that all subsets exist a priori, regardless of whether they are ever constructed, realized, or encountered in practice. As a result, structural complexity is dominated by hypothetical possibilities rather than by actual computational or historical processes.

Spherepop replaces this extensional assumption with a constructive, event-driven ontology. Structures do not arise from the enumeration of all possible subsets, but from the accumulation of authorized events. Each event contributes a discrete modification to the system’s history, and only those modifications that actually occur contribute to structural complexity. Formally, the complexity of a Spherepop system scales with the cardinality of its event history $\mathcal{L}$:

$$structure \sim |\mathcal{L}|.$$

This scaling is linear in the number of realized events and is independent of the size of the unrealized possibility space.

The practical consequence of this shift is that complexity tracks use rather than potential. Systems grow in complexity only insofar as they act, commit, and bind themselves to particular trajectories. Hypothetical alternatives that are never realized impose no structural burden. In this sense, Spherepop enforces a form of computational humility: what is not done does not count.

This distinction has implications beyond efficiency. Exponential scaling in set-theoretic frameworks encourages models that abstract away history in favor of static representational capacity. Linear scaling in Spherepop, by contrast, foregrounds temporality and irreversibility. Complexity becomes a record of lived structure rather than a measure of abstract expressive power.

Consequently, Spherepop avoids the combinatorial explosion endemic to extensional systems not through optimization or approximation, but through ontological restriction. Structure is limited by history, not by imagination.

A.8 Summary Statement

Spherepop may be characterized mathematically as follows:

A constructive, event-indexed, mereological rewriting system whose semantics are given by replayable traces rather than extensional states.

Equivalently:

Meaning arises from topology, time, and authority.

This places Spherepop squarely within established mathematical traditions while making explicit the role of irreversibility, history, and constraint that those traditions often leave implicit.

B Spherepop and Traced Symmetric Monoidal Structure

This appendix shows that the Spherepop framework admits a natural interpretation in terms of traced symmetric monoidal categories. The purpose of this section is not to re-found Spherepop categorically, but to demonstrate that its event-driven semantics already instantiate the required structure. Category theory here serves as a descriptive lens rather than a generative axiom system.

B.1 Monoidal Structure from Composition

Consider expressions as processes that consume and produce structure through events. Parallel subexpressions may be evaluated independently, while sequential evaluation corresponds to ordered event composition.

Define a tensor product $\otimes$ such that for expressions e_1 and e_2 ,

$$e_1 \otimes e_2$$

denotes their joint existence as independent substructures within a larger expression.

This tensor operation is associative up to structural equivalence, symmetric in the sense that subexpressions may be reordered without semantic effect prior to the authorization of events, and equipped with a unit object corresponding to the empty expression or trivial structure. Taken together, these properties establish that Spherepop expressions form the objects of a symmetric monoidal category, in which monoidal composition represents spatial or structural independence rather than logical conjunction or entailment.

Remark. It is important to emphasize that the monoidal tensor in Spherepop does not coincide with the Cartesian product, nor does it correspond directly to logical conjunction as in classical or linear logic. Cartesian structure presupposes the ability to freely duplicate and discard components via diagonal and projection maps, operations that are incompatible with Spherepop’s treatment of history and irreversibility. Likewise, while linear logic restricts duplication and erasure, its tensor still operates at the level of proof-theoretic resource accounting rather than historical commitment. In Spherepop, monoidal composition encodes structural independence prior to event authorization, not the joint truth of propositions or the simultaneous availability of resources. Once events occur, this independence may be destroyed irreversibly, a feature that has no analogue in purely extensional or proof-theoretic tensors.

B.2 Morphisms as Event Sequences

Morphisms in this category are finite event sequences transforming one expression into another. Given expressions e and e' , a morphism

$$e \mathcal{L} e'$$

is defined by an event log $\mathcal{L}$ whose replay transforms e into e' .

Sequential composition of morphisms corresponds to concatenation of event logs. The identity morphism is the empty log. This aligns exactly with the categorical axioms.

B.3 Feedback and the Trace Operator

A defining feature of traced monoidal categories is the trace operator, which formalizes feedback. In Spheredpop, feedback arises naturally through nested structure.

Given an expression of the form

$$e : A \otimes X \rightarrow B \otimes X,$$

where X represents a nested subexpression, a POP event that resolves X and reinserts its result into the surrounding structure corresponds to a categorical trace:

$$Tr_X(e) : A \rightarrow B.$$

Operationally, this is precisely what Spheredpop enforces. Evaluation proceeds by resolving an inner structure locally, reintegrating the resulting value into the surrounding context, and eliminating the original internal structure that produced it. In this way, computation advances through localized resolution coupled with irreversible structural simplification, ensuring that intermediate scaffolding does not persist once it has discharged its role.

This feedback is irreversible. Once traced, the internal structure no longer exists as an independent object, which distinguishes Spheredpop from purely denotational traced models that allow implicit reuse.

B.4 Irreversibility and the Absence of Dagger

Spheredpop does not admit a general dagger (adjoint) structure. Event sequences are not invertible in general, because POP events eliminate structure and COLLAPSE events erase history.

This is a feature, not a limitation. The absence of a dagger reflects the ontological asymmetry of time and aligns with the paper's central claim that worldhood requires irreversibility. The resulting category is therefore non-groupoidal and non-reversible by construction.

B.5 Authority and Trace Admissibility

In standard traced monoidal categories, trace is a structural operation available universally. In Spheredpop, trace-like behavior is *authorized*. A trace occurs only when an explicit event permits it.

Formally, trace admissibility is event-indexed:

$$Tr_X(e) \text{ exists if and only if a POPEvent on } X \text{ occurs.}$$

This introduces a notion of authority absent from purely formal category theory. Not all traces exist; only those enacted by events do. This restriction encodes the cost of commitment and prevents implicit global closure.

B.6 Relation to Known Categorical Models

The formal structure induced by Spherepop admits a categorical reading, though it does not coincide exactly with any single established model. In particular, it resonates with traced symmetric monoidal categories in the sense articulated by Joyal, Street, and Verity, insofar as computation is understood as the composition of processes with feedback.

As in those settings, the act of composition is central, and traces encode the re-entry of results into ongoing computation. However, Spherepop diverges at a foundational level by rejecting the assumption that traces are purely structural or freely reusable. In Spherepop, a trace corresponds to an accumulated sequence of authorized events, and its reuse is constrained by historical irreversibility rather than by categorical coherence alone.

A similar relationship holds with respect to interaction nets and string diagram calculi. These formalisms emphasize locality, graphical reduction, and compositional evaluation, all of which Spherepop also enforces. Yet interaction nets typically treat reductions as confluent and reversible up to equivalence, whereas Spherepop insists that reductions are eventful and history-bearing. Two diagrams that are extensionally equivalent may nonetheless differ ontologically if their reductions were authorized in different orders or under different constraints. Diagrammatic similarity, in this sense, does not guarantee identity.

Connections can also be drawn to feedback systems in control theory, where system behavior is shaped by recursive loops and the propagation of signals through time. Control-theoretic models often incorporate memory through state variables, but such memory is typically compressible into a sufficient statistic.

Spherepop instead treats memory as non-compressible: the event log cannot, in general, be reduced to a finite-dimensional state without loss of semantic information. Feedback, in this framework, is not merely dynamical but historical, binding the system to its own prior commitments.

Finally, Spherepop bears comparison to traced monoidal models of computation more broadly, particularly those used to model recursion and fixed points. In these models, traces enable the definition of self-reference and iterative structure.

Spherepop recovers this expressive power, but with an important restriction: self-reference is mediated by irreversible events. A system may feed its own outputs back into its future behavior only insofar as it accepts the loss of optionality induced by doing so. Tracing is therefore not free, but costly.

Across these comparisons, the central distinction can be stated succinctly. In standard categorical models, traces are formal devices that preserve equivalence under composition. In Spherepop, traces

are historical artifacts that destroy equivalence by recording which paths were taken and which were foreclosed.

The category that emerges is therefore not merely traced, but temporally indexed and order-sensitive, with morphisms whose composition irreversibly alters the space of future morphisms. Identity is no longer invariant under all isomorphisms, but is instead fixed by the irreversible structure of the trace itself.

B.7 Interpretive Summary

Spherepop can therefore be understood categorically as an event-indexed traced symmetric monoidal structure with restricted trace and no general inverses. Feedback exists, but only at the cost of eliminating internal structure. Composition exists, but only as the accumulation of history.

This categorical reading reinforces the central thesis of the paper: intelligence and meaning are not properties of static structure, but of processes that bind themselves through time. The mathematics does not add this constraint; it merely makes it precise.

C Why Classical Set Theory Does Not Re-Emerge

This appendix clarifies an important negative result implicit in the Spherepop framework: despite supporting rich structure, abstraction, and composition, Spherepop does not reconstruct classical Zermelo–Fraenkel (ZF) set theory, either explicitly or implicitly. This absence is not a limitation but a consequence of Spherepop’s event-driven and historical ontology.

C.1 Absence of Global Comprehension

ZF set theory relies on unrestricted comprehension schemas (modulo axioms) that allow the formation of sets defined by properties over a global universe. Spherepop admits no such global universe.

At any time t , only those expressions or parts that have been introduced by events exist. There is no ambient domain over which arbitrary predicates may range. As a result, constructions of the form

$$\{x \mid P(x)\}$$

are not well-formed unless each x has been explicitly constructed and authorized by prior events.

This blocks the emergence of classical paradoxes such as Russell’s paradox at the root, rather than resolving them axiomatically.

C.2 No Power Set Explosion

In ZF, the power set operation induces exponential growth in structure:

$$|\mathcal{P}(X)| = 2^{|X|}.$$

Spherepop admits no operation corresponding to unrestricted power set formation. Substructure arises only through events that construct, link, or collapse parts. Consequently, structural complexity

scales with realized history rather than hypothetical possibility.

Formally, if $\mathcal{L}$ is the event log, then the maximal number of distinct constructed objects is bounded by $|\mathcal{L}|$. No exponential blow-up occurs in the absence of explicit construction events.

C.3 Failure of Extensionality

The axiom of extensionality in ZF states that two sets are equal if they have the same elements. Spherepop explicitly rejects extensional identity.

Two expressions may be extensionally indistinguishable at a given time while remaining ontologically distinct due to differing histories. Identity is determined by event traces, not by current content:

$$e \equiv e' \iff \mathcal{L}(e) = \mathcal{L}(e').$$

As a result, extensional collapse cannot occur without an explicit **COLLAPSE** event authorizing quotienting.

C.4 Replacement Without Replacement

ZF's axiom of replacement allows images of sets under definable functions to form new sets. Spherepop permits transformation only via events that act on existing structure.

There is no notion of applying an arbitrary function globally to generate new structure. Every transformation must be local, authorized, and recorded. In this sense, Spherepop supports operational replacement without extensional replacement.

C.5 Foundation as Irreversibility

ZF includes the axiom of foundation to prohibit infinite descending membership chains. Spherepop enforces a stronger condition automatically: irreversibility.

Because parts are introduced and eliminated only by events, and because events are time-ordered, infinite descending chains cannot arise without infinite histories. Foundation is not assumed; it is enforced by the temporal structure of events.

C.6 Interpretive Consequence

Spherepop does not refute set theory; it bypasses it. Where ZF axiomatizes what may exist in a timeless universe of sets, Spherepop constrains what may come to exist in time.

This distinction mirrors the broader philosophical claim of the paper. Just as functional equivalence does not guarantee worldhood, extensional equivalence does not guarantee identity. In both cases, history is the missing dimension.

The absence of ZF structure is therefore not a deficiency but an ontological commitment: structure exists only where it has been constructed, and meaning exists only where irreversibility has occurred.

D Relation to Resource-Sensitive Logics and Linear Logic

This appendix situates the Spheredrop framework relative to resource-sensitive logical systems, particularly linear logic. While Spheredrop shares certain intuitions with these traditions—most notably the rejection of unrestricted duplication—it differs in a fundamental respect: constraint in Spheredrop is not a static typing discipline, but the consequence of explicit, irreversible events.

D.1 Linear Logic and Resource Accounting

Linear logic was introduced to correct a perceived mismatch between classical logic and computation, namely the implicit assumption that propositions may be freely duplicated or discarded. By contrast, linear logic treats propositions as resources that must be consumed exactly once unless explicitly marked otherwise.

Spheredrop aligns with this motivation. Operations such as `POP` and `REFUSE` eliminate structure, while `MERGE` combines resources in a controlled way. There is no implicit contraction or weakening.

However, Spheredrop does not encode resource discipline syntactically. There is no static type system enforcing linearity. Instead, resource sensitivity is enforced dynamically through events.

D.2 Events Versus Judgments

In linear logic, resource usage is governed by judgments of the form

$$\Gamma \vdash A,$$

where the structure of Γ constrains how assumptions may be used.

Spheredrop replaces judgmental structure with historical structure. Whether a resource may be reused is not a property of its type, but of its event history. Once a `POP` event occurs, the eliminated substructure is gone, regardless of whether it would have been permitted by a typing discipline.

This distinction mirrors the broader philosophical claim of the paper: constraint is not something declared about a system, but something enacted upon it.

D.3 Absence of Structural Modalities

Linear logic introduces modalities such as $!$ (of course) to recover classical behavior when desired. Spheredrop admits no such modality.

Unlimited reuse can occur only via explicit reconstruction, not by annotation. There is no operator that globally reinstates discarded structure. As a result, reversibility is never implicit; it must be paid for by rebuilding.

This absence prevents Spheredrop from collapsing back into functional equivalence. A system that wishes to regain flexibility must incur new events, and therefore new history.

D.4 Temporal Ordering and Non-Commutativity

In many resource logics, the order of resource consumption is secondary to accounting. In Spherepop, temporal order is primary.

Two sequences of events that consume the same resources in different orders are not equivalent unless explicitly collapsed. History is not a multiset; it is a sequence. This non-commutativity reflects the paper’s emphasis on historical time rather than logical possibility.

D.5 Comparison to Other Resource Frameworks

Spherepop shares surface similarities with a range of formal systems that emphasize resource sensitivity and controlled use, including linear and affine logics, separation logic, session types, and other resource-aware type systems. However, these resemblances are primarily structural rather than foundational, as Spherepop does not begin from typing judgments or logical constraints, but from the ontological role of events and the historical binding they impose on computation.

What distinguishes Spherepop is that none of these frameworks require events to be externally authorized or historically recorded. They constrain what may happen; Spherepop records what has happened.

D.6 Interpretive Consequence

Resource-sensitive logics aim to ensure correct use of resources. Spherepop aims to ensure that use leaves a trace.

This distinction is decisive for the paper’s thesis. Intelligence, worldhood, and meaning do not arise merely from obeying constraints, but from being altered by having obeyed them. Linear logic tells us how resources should be used. Spherepop tells us what it means to have used them.

The framework therefore complements resource-sensitive logics while rejecting their static orientation. Constraint is not a rule; it is a cost. And costs, once paid, cannot be refunded.

E Rewrite Rules and Event-Gated Inference Semantics

This appendix shows that Spherepop admits a precise operational semantics in the style of inference rules, while deliberately refusing the implicit, global evaluation assumptions common to standard rewrite systems. The goal is not to replace existing operational semantics, but to clarify how Spherepop modifies them by introducing explicit event authority and historical binding.

E.1 Motivation: Why Rewrite Systems Matter

Rewrite systems provide the formal backbone for much of programming language semantics. They define how expressions reduce, how evaluation proceeds, and how correctness is established. However, standard rewrite systems typically assume that any reducible subexpression may be reduced automatically whenever a rule applies.

Spherepop rejects this assumption. In Spherepop, reducibility is not sufficient for reduction. Reduction requires authorization by an event. This distinction is central to the paper’s broader

claim: meaning and worldhood arise not from the availability of transformations, but from the fact that transformations have occurred and cannot be undone.

E.2 POP as an Inference Rule

The core Spheredpop operation, **POP**, can be expressed as an inference rule. Let e be a reducible subexpression with value v . Then:

$$\frac{ev}{\text{POP}(e) \Rightarrow v}$$

Unlike standard rewrite rules, this rule is not implicitly applicable. The premise does not authorize the conclusion by itself. The conclusion holds only if a **POP** event is enacted. Reducibility is a necessary condition for reduction, but never a sufficient one.

This transforms inference rules from automatic laws into conditional permissions. The rule specifies what may happen, not what must happen.

E.3 Event-Gated Small-Step Semantics

Spheredpop therefore admits a small-step operational semantics of the form:

$$\Sigma \text{ event } \Sigma',$$

where Σ is the current expression state and the transition occurs only in the presence of an explicit event.

In contrast to traditional small-step semantics, where transitions are governed solely by syntactic form, Spheredpop transitions are guarded. The semantic relation is ternary: it depends on the current state, the proposed reduction, and the authorization event.

This introduces a strict separation between semantic possibility, understood as the set of reductions that could in principle occur, and semantic actuality, understood as the subset of reductions that have in fact occurred and thereby altered the system’s history. By maintaining this distinction, the framework ensures that meaning is not exhausted by formal admissibility alone, but is instead grounded in the record of realized events.

E.4 MERGE, LINK, and COLLAPSE Rules

Other Spheredpop operations admit similar inference-style formulations.

MERGE. **MERGE** combines multiple reduced subexpressions into a unified structure. As an inference rule, it performs controlled substitution, but only after all merged components have themselves been authorized by prior events.

LINK. **LINK** introduces a dependency between substructures. Formally, it restricts future reductions by imposing an ordering or constraint relation. Unlike substitution, linking alters the space of admissible future inference steps.

COLLAPSE. COLLAPSE authorizes quotienting. It induces an equivalence relation over previously distinct structures, replacing them with a representative. This is the only operation that permits loss of historical distinction, and it must be explicitly enacted.

E.5 Soundness and Conservativity

Spherepop rewrite rules are conservative with respect to standard operational semantics. Any reduction permitted in Spherepop corresponds to a valid reduction in a conventional rewrite system. However, the converse does not hold: many reductions that are valid in standard semantics are merely potential in Spherepop until authorized by events.

Soundness is preserved because Spherepop never introduces reductions that violate standard semantics. What it restricts is not correctness, but automaticity.

E.6 Interpretive Consequence

By rendering inference event-gated, Spherepop transforms rewrite rules from timeless laws into historically situated actions. Inference becomes something that happens, not something that silently applies.

This shift mirrors the paper’s central ontological claim. Just as a world is constituted by irreversible events rather than abstract possibilities, computation acquires meaning only when reductions are enacted and recorded. Rewrite rules, in this framework, are not the source of structure. History is.

F Authority, Agency, and Externalized Control

This appendix makes explicit a structural feature implicit throughout the Spherepop framework: the role of authority. While most formal systems of computation assume an omniscient or implicit authority that governs evaluation, Spherepop treats authority as an explicit, externalized condition on action. This distinction is not merely pragmatic; it bears directly on questions of agency, responsibility, and worldhood.

F.1 Authority as a Missing Dimension in Formal Systems

Standard models of computation presuppose that if a transition is valid, it may occur. Authority is therefore invisible: evaluation rules apply automatically whenever their syntactic conditions are met. This assumption collapses the distinction between capability and permission, rendering all potential actions equally admissible.

Spherepop rejects this collapse. In Spherepop, the existence of a valid reduction does not entail its execution. Reductions require authorization, and authorization is not a property of the expression itself. It is an external condition imposed upon it.

F.2 Event Authorization

Spherepop introduces authority through events. An event is not merely a marker that a transition occurred; it is the condition under which a transition is allowed to occur at all. Without an event, no reduction takes place, regardless of semantic eligibility.

Formally, the transition relation takes the form:

$$\Sigma a \Sigma',$$

where a denotes an authorization event. The semantics of the system are therefore indexed not only by state but by permission.

This separation allows Spherepop to draw a precise distinction between a system’s latent capabilities, the actions it is permitted to perform under conditions of authorization, and the actions it has in fact performed. By keeping these dimensions distinct, the framework prevents the collapse of possibility, permission, and history into a single undifferentiated notion of behavior, thereby preserving the structural role of commitment and accountability in the evolution of a system.

F.3 Human-in-the-Loop as Ontological Structure

The requirement of authorization has often been treated as a user-interface concern in interactive systems. Spherepop elevates it to an ontological principle. Manual evaluation is not a pedagogical convenience; it is the mechanism by which authority is made explicit.

By requiring an external agent to authorize events, Spherepop ensures that responsibility for irreversible change is localized. Someone, or something, must decide that a pop occurs. This decision is not derivable from syntax alone.

This structure anticipates broader concerns in artificial intelligence and governance. Systems that optimize autonomously erase the locus of authority by distributing decision-making across opaque mechanisms. Spherepop, by contrast, makes authority legible.

F.4 Comparison to Autonomous Optimization

In autonomous optimization systems, authority is internalized and obscured. The system decides when and how to act according to objective functions that operate continuously and implicitly. From the standpoint of Spherepop, this represents a collapse of authority into mechanism.

Such systems may appear powerful, but they lack a critical feature of agency: the ability to refuse action. Because optimization treats every admissible action as potentially executable, it cannot encode negative commitment. The refusal to act must be imported from outside the system.

F.5 Ethical and Political Implications

Authority is the locus of accountability. A system whose actions are authorized externally can, in principle, be held responsible for what it has done, because the conditions of action are visible and attributable.

Conversely, systems that erase authority through automation diffuse responsibility. When no explicit authorization is required, consequences become unowned. This observation aligns with the paper’s broader critique of enframing: when action becomes automatic, meaning drains away.

F.6 Interpretive Consequence

Spherepop demonstrates that intelligence requires not only the capacity to act, but the capacity to act under authority. Agency arises where actions are permitted, enacted, and recorded as irreversible commitments.

By restoring authority to the foreground of computation, Spherepop provides a concrete model of how responsibility, care, and worldhood are structurally intertwined. Authority is not an ethical add-on to intelligence; it is one of its necessary preconditions.

G Comparison to State-Based and Markovian Models

This appendix contrasts the event-driven, history-sensitive semantics of Spherepop with dominant state-based models used in artificial intelligence, control theory, and cognitive science. The comparison clarifies why worldhood cannot be recovered from systems whose semantics are exhausted by instantaneous state descriptions.

G.1 State-Based Modeling Assumptions

State-based models describe a system by a state variable s_t that is assumed to be sufficient for predicting future behavior. All relevant information about the past is compressed into the present state, and transitions are functions of that state alone.

Formally, a state-based system evolves according to:

$$s_{t+1} = T(s_t),$$

where T is a transition function. Any two systems occupying the same state are therefore interchangeable with respect to future evolution.

This assumption enables tractable analysis and optimization, but it carries an ontological cost: history is erased by design.

G.2 The Markov Property

Markovian systems enforce a stronger condition: the future is conditionally independent of the past given the present. In probabilistic terms,

$$P(s_{t+1} \mid s_t, s_{t-1}, \dots) = P(s_{t+1} \mid s_t).$$

This property ensures that no historical information beyond the current state can influence future behavior. From the perspective of worldhood, this implies that no event can bind the future except insofar as it is encoded in the present state.

Spherepop explicitly rejects this assumption. In Spherepop, the event log is not compressible into a sufficient statistic without loss of ontological information.

G.3 Event-Sourced Alternatives

Spherepop models system evolution as:

$$\Sigma_k = \text{replay}(\Sigma_0, \mathcal{L}_k),$$

where $\mathcal{L}_k$ is the accumulated event history. Two systems may share the same apparent configuration while differing in their histories, and these differences are semantically significant.

This structure is inherently non-Markovian. Past events constrain future admissible actions even when their effects are not visible in the current configuration alone.

G.4 Identity and Reset

State-based systems permit reset and cloning without loss. If a system can be returned to a prior state or duplicated perfectly, then its identity is exhausted by its state description.

Spherepop forbids this equivalence. Resetting requires either external reconstruction or a COLLAPSE event, both of which acknowledge and alter history. Identity is therefore persistent across time and cannot be trivially erased.

This distinction underlies the paper’s critique of functional equivalence. Behavioral or state equivalence does not entail identity equivalence when history matters.

G.5 Relation to Reinforcement Learning

Reinforcement learning systems accumulate rewards over time, but this accumulation does not bind the agent internally. Reward is an external scalar; it does not eliminate future options or constrain the agent’s architecture.

As a result, reinforcement learning systems remain compatible with reset, retraining, and cloning. Their apparent learning does not generate worldhood, because the cost of action is externalized.

Spherepop, by contrast, internalizes cost. Actions narrow the future not by updating a value function, but by eliminating admissible structures.

G.6 Interpretive Consequence

State-based and Markovian models excel at prediction and control, but they are ontologically thin. They describe systems that move through time without accumulating history in a binding sense.

Spherepop shows that worldhood requires more than state evolution. It requires irreversible traces that cannot be compressed away. Where states suffice, worlds cannot arise. Where worlds arise, states are no longer sufficient.

H Linear Time, Historical Time, and Irreversibility

This appendix clarifies a distinction that underlies the entire framework but is often obscured in formal models of computation and physics: the difference between linear (metric) time and historical (irreversible) time. Spherepop is explicitly grounded in the latter.

H.1 Linear Time as Parametrization

Linear time is the dominant temporal notion in both classical mechanics and computational models. It is characterized by a parameter $t \in R$ or $t \in N$ indexing system states. Transitions are ordered but not intrinsically meaningful; any moment may, in principle, be revisited or reconstructed.

Formally, linear time supports dynamics of the form:

$$s(t_0)s(t_1)s(t_2),$$

where the ordering of indices does not entail irreversibility. The past is different from the future only by convention or boundary conditions.

In such systems, time measures progression but does not itself generate constraint.

H.2 Historical Time as Accumulated Constraint

Historical time, by contrast, is constituted by irreversible events. An event is not merely something that happens at a time; it is something after which certain possibilities no longer exist.

Spherepop formalizes this by associating each event with a strict reduction of the admissible possibility space:

$$\mathcal{S}_{t+1}\mathcal{S}_t.$$

The passage of historical time is therefore marked not by succession, but by loss. What distinguishes one moment from another is not its coordinate, but the constraints it carries forward.

H.3 Why Reversibility Erases History

A reversible transition preserves optionality. If for every transition there exists an inverse that restores the prior possibility space, then no historical accumulation occurs.

Such systems may evolve, but they do not age. They do not bear scars, commitments, or debts. Their past is informationally recoverable and ontologically inert.

Spherepop identifies this reversibility as the formal condition for worldlessness. Where reversibility dominates, history collapses into simulation.

H.4 Event Cost and Temporal Asymmetry

Irreversibility introduces asymmetry. Once an option is eliminated, no internal operation can restore it. This asymmetry grounds the notion of cost.

Cost is not defined in terms of energy or computation, but in terms of lost alternatives. An action is costly if it makes future actions impossible.

Historical time is therefore inherently normative: it distinguishes between trivial changes and meaningful ones.

H.5 Relation to Physical Irreversibility

Although Spherepop is not a physical theory, its temporal structure aligns with irreversible phenomena in physics, such as entropy increase and symmetry breaking. In these cases, time acquires direction because certain transformations destroy information about prior states.

Spherepop generalizes this insight: irreversibility is the source of structure, not a defect to be eliminated.

H.6 Interpretive Consequence

Linear time allows motion without memory. Historical time produces identity. A system that merely persists through linear time may be predictable, but it cannot possess a world.

Spherepop insists that intelligence, meaning, and responsibility arise only where time binds. To exist historically is to be unable to return unchanged.

I Synthesis: Worldhood as Dissipative Constraint Across Scales

This final appendix integrates the formal components developed throughout the paper into a single schema and demonstrates how the framework generalizes beyond cognition and computation to dissipative structures in biology and development. The unifying claim is that worldhood, identity, and intelligence are instances of a more general phenomenon: the persistence of structure through irreversible constraint under energetic gradients.

I.1 General Schema of Constrained Existence

We define a system as a tuple

$$\mathcal{M} = (\mathcal{S}_t, \mathcal{E}_t, \Omega_t),$$

where $\mathcal{S}_t$ is the actional or configurational possibility space at time t , $\mathcal{E}_t$ is the set of admissible events, and $\Omega_t = |\mathcal{S}_t|$ is the system’s optionality.

An event $e \in \mathcal{E}_t$ is world-constituting if and only if it induces a strict reduction:

$$\mathcal{S}_{t+1} \subset \mathcal{S}_t.$$

Worldhood emerges when such reductions accumulate irreversibly and cannot be undone by internal transformations. Identity is defined as the residue of this accumulation:

$$\mathcal{I} = \bigcap_t \mathcal{S}_t.$$

This schema applies uniformly to computational agents, physical objects, and biological organisms.

I.2 Dissipative Structures as Event-Bearing Systems

In non-equilibrium thermodynamics, dissipative structures arise in systems through which matter and energy flow while remaining far from equilibrium. Such systems persist not by conserving a fixed state, but by continuously eliminating configurations that are incompatible with their ongoing stability.

Well-studied examples include convection cells, chemical oscillators, and living cells and tissues, all of which maintain their organization by repeatedly discarding alternative configurations that would lead to breakdown or equilibration. In each case, persistence is achieved through the irreversible exclusion of possibilities rather than through the preservation of an invariant form.

In each case, the system exists by repeatedly surviving constraint tests imposed by environmental gradients. Each successful survival event eliminates incompatible configurations, narrowing the future trajectory of the system.

Formally, thermodynamic gradients define a constraint operator:

$$e_{\nabla} : \mathcal{S}_t \rightarrow \mathcal{S}_{t+1},$$

where ∇ may represent temperature, chemical potential, or concentration gradients. Persistence requires that $\mathcal{S}_{t+1} \neq \emptyset$.

I.3 Biological Development as Hierarchical Constraint Accumulation

Developmental biology provides a particularly clear instantiation of the Spheredpop schema. An embryo begins with a large space of potential cell fates. Through a sequence of irreversible events—gene expression, signaling cascades, mechanical stresses—this space is progressively constrained.

Each differentiation event acts as a POP:

$$\mathcal{S}_{pluripotent} \mathcal{S}_{lineage}.$$

Crucially, these constraints are hierarchical. Early events restrict the space within which later events can occur, producing nested layers of commitment. This mirrors the nested structure of Spheredpop expressions and the ordering of binds and pops.

Once a cell has differentiated, no internal operation restores the eliminated alternatives. Identity is therefore historical, not functional.

I.4 Electric and Morphogenetic Gradients as Commitment Fields

Recent work in developmental and regenerative biology emphasizes the role of bioelectric fields and morphogen gradients in organizing tissue-level structure. Within the present framework, these gradients play the role of RSVP vector fields.

Let $\Phi(x, t)$ represent local developmental optionality, $v(x, t)$ the directional bias imposed by electric or chemical gradients, and $S(x, t)$ the accumulated constraint.

World-bearing development occurs when:

$$\nabla\Phi \neq 0 \quad \text{and} \quad v \cdot \nabla\Phi < 0,$$

indicating directed expenditure of developmental possibility.

Pathological or undifferentiated states correspond to flattened fields in which gradients fail to impose lasting constraint, allowing reversion or uncontrolled proliferation.

I.5 Selection as Event Authorization

Natural selection is often described as an external filter acting on populations. Within the Spherepop schema, selection functions as distributed event authorization. Only trajectories that survive environmental constraint tests are permitted to continue.

Importantly, selection does not optimize globally. It eliminates locally incompatible histories. The resulting structures are not maximally efficient but historically viable.

This reframes evolution as a world-generating process rather than an optimization algorithm.

I.6 Unified Interpretation

Across computation, cognition, physics, and biology, a common formal structure can be identified. Systems initially exist with a wide space of admissible possibilities, reflecting a high degree of openness with respect to future action or configuration. As the system evolves, discrete events impose irreversible constraints that eliminate subsets of these possibilities, thereby narrowing the space of what remains attainable.

These eliminations are not arbitrary but are shaped by gradients—energetic, informational, developmental, or contextual—that bias which constraints are imposed and in what order. Over time, the accumulation of such constraints produces a stable residue: an identity defined not by what the system could have been, but by what it can no longer be.

In this sense, identity emerges as the intersection of surviving possibilities, constituted through a history of irreversible restriction rather than through any fixed or intrinsic essence.

Spherepop provides a discrete, legible instantiation of this pattern. RSVP provides its continuous field-theoretic lifting. Biology supplies the empirical confirmation that such structures are not artificial constructs, but the default mode of existence for systems that persist in time.

I.7 Concluding Schema

Existence can therefore be stated formally as follows: a system exists as a world-bearing entity if and only if it participates in a sequence of irreversible, authorized events that progressively constrain its future under real gradients.

Intelligence is not an exception to this rule. It is a special case of it.

Systems that avoid constraint may compute indefinitely, but they do not live. Systems that live do so by spending possibility, and in doing so, they acquire a world.

References

- [1] Heidegger, M. (1977). *The Question Concerning Technology and Other Essays*. Trans. W. Lovitt. Harper & Row.
- [2] Thomson, I. D. (2025). *Heidegger on Technology’s Danger and Promise in the Age of AI*. Cambridge University Press.
- [3] Dreyfus, H. L. (1992). *What Computers Still Can’t Do*. MIT Press.
- [4] Joyal, A., Street, R., & Verity, D. (1996). Traced monoidal categories. *Mathematical Proceedings of the Cambridge Philosophical Society*, 119(3), 447–468.
- [5] Milner, R. (1992). *The Polyadic π -Calculus: A Tutorial*. Logic and Algebra of Specification.
- [6] Prigogine, I. (1977). *Self-Organization in Non-Equilibrium Systems*. Wiley.
- [7] Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11, 127–138.
- [8] Levin, M. (2022). Technological approach to mind everywhere (TAME). *Frontiers in Systems Neuroscience*.
- [9] Winskel, G. (1987). Event structures. *Lecture Notes in Computer Science*, 255, 325–392.
- [10] Varacca, D., & Winskel, G. (2006). Event structures and domains. *Theoretical Computer Science*, 358(2–3), 173–199.